## Preference Loop: Personalized Controller Optimization

Scenario $s$ (initial velocity, bump geometry), controller parameter $c \in \mathcal{C} = [30, 300]^P$, trajectory $\tau(c, s) = \mathrm{rollout}(c, s)$, episode feature $z = \phi(\tau) \in \mathbb{R}^D$, user preference $\boldsymbol{\theta}_i \in \mathbb{R}^D$. The user-optimal controller is

$$c_i^* = \arg\max_c \ \boldsymbol{\theta}_i^\top g(c), \qquad g(c) := \mathbb{E}_s[z(c, s)]\,,$$

and the loop replaces both unknowns with estimates: $\boldsymbol{\theta}_i \to \bar{\boldsymbol{\theta}}_i$ (hierarchical Bayes posterior mean) and $g \to \hat{g}$ (online: fixed-scenario Monte Carlo; offline: scenario-covariate regression).

**Algorithm 1** Preference data generation and reward estimation

**for** user $i = 1, \dots, n$ **do**
  **for** episode $j = 1, \dots, N$ **do**
    $c_{ij} \sim U(\mathcal{C}), \qquad s_{ij} \sim p(s)$
    $\tau_{ij} \leftarrow \mathrm{rollout}(c_{ij}, s_{ij})$
    $z_{ij} \leftarrow \phi(\tau_{ij})$
    $y_{ij} \sim \mathrm{Bernoulli}\big(\sigma(\boldsymbol{\theta}_i^\top z_{ij})\big)$
  **end for**
**end for**
$\{\bar{\boldsymbol{\theta}}_i\}_{i=1}^n \leftarrow \mathrm{HierBayesFit}\big(\{(z_{ij}, y_{ij})\}\big)$
$S_{\mathrm{eval}} \leftarrow \{s'_1, \dots, s'_K\}$ {held-out scenarios}

**Algorithm 2** Search subroutines (return a maximizer of $J$)

**function** $\mathrm{Grid}(J)$
  $\mathcal{G} \leftarrow$ linspace grid over $\mathcal{C}$ $\{20^P$ candidates$\}$
  **return** $\arg\max_{c \in \mathcal{G}} J(c)$

**function** $\mathrm{CMAES}(J)$ {normalized $[0, 1]^P$ space}
  $m \leftarrow 0.5, \quad \sigma \leftarrow 0.25, \quad C \leftarrow I$
  **for** generation $= 1, \dots, 15$ **do**
    sample $x_1, \dots, x_8 \sim \mathcal{N}(m, \sigma^2 C)$
    decode each $x_k$ into $\mathcal{C}$, score $J(x_k)$
    update $(m, \sigma, C)$ toward top-ranked $x_k$
  **return** $\mathrm{decode}(m)$

**Algorithm 3** Online optimization

$S_{\mathrm{opt}} \leftarrow \{s_1, \dots, s_M\}$
**for** user $i = 1, \dots, n$ **do**
  $\hat{J}_i(c) := \mathbb{E}_{\theta_i \mid \mathcal{D}}\Big[\frac{1}{M} \sum_{s \in S_{\mathrm{opt}}} \theta_i^\top z(c, s)\Big]$ {linear reward}
  $\quad = \frac{1}{M} \sum_{s \in S_{\mathrm{opt}}} \bar{\boldsymbol{\theta}}_i^\top z(c, s)$
  $\hat{c}_i \leftarrow \mathrm{Search}(\hat{J}_i)$ $\{\mathrm{Search} \in \{\mathrm{Grid}, \mathrm{CMAES}\}\}$
**end for**
**return** $\{\hat{c}_i\}_{i=1}^n, \qquad \hat{c}_i = \arg\max_{c \in \mathcal{C}} \hat{J}_i(c)$

**Algorithm 4** Offline optimization and evaluation

$\mathcal{D} \leftarrow \bigcup_{i \in \mathrm{train}} \{(c_{ij}, s_{ij}, z_{ij})\}_{j=1}^N$ {pooled log, no rollouts}
$\hat{f} \leftarrow \mathrm{fit}\big((c, s) \mapsto z;\ \mathcal{D}\big)$ {polynomial ridge}
$\hat{g}(c) := \frac{1}{n} \sum_j \hat{f}(c, s_j)$ {logged scenarios = test course}
**for** user $i = 1, \dots, n$ **do**
  $\hat{J}_i(c) := \bar{\boldsymbol{\theta}}_i^\top \hat{g}(c), \qquad J_i^*(c) := \boldsymbol{\theta}_i^\top \hat{g}(c)$
  $\hat{c}_i \leftarrow \mathrm{Search}(\hat{J}_i)$ {no rollouts}
  $c_i^{\mathrm{oracle}} \leftarrow \mathrm{Search}(J_i^*)$
  $J_i^{\mathrm{eval}}(c) := \frac{1}{K} \sum_{s \in S_{\mathrm{eval}}} \boldsymbol{\theta}_i^\top z(c, s)$ {simulator only here}
  $\mathrm{regret}_i \leftarrow J_i^{\mathrm{eval}}(c_i^{\mathrm{oracle}}) - J_i^{\mathrm{eval}}(\hat{c}_i)$
**end for**